

1. Big picture: “Experimental setups”

This week is about **techniques** you can use during a project to:

- Discover user needs
- Improve the design
- Test the product before and after release

Your slide list of techniques:

- Ethnographic research
- Participatory design
- Focus group research
- Surveys
- Walkthroughs
- Open and closed card sorting
- Paper prototyping
- Expert / heuristic evaluations
- Usability testing
- Follow-up studies

👉 **MEMORIZE this list.** Teacher can easily say “Name 5 experimental setups/methods to build in usability.”

We’ll go through each.

2. Ethnographic research

What it is:

A **qualitative** method where you **observe users in their real environment** – where they actually use (or would use) the product.

You don't bring them to the lab; you go to **their world**:

- In their office
- On the shop floor
- In their classroom
- In their home

Goal:

- Understand the **target users**, their **tasks**, and their **goals** in rich detail
- Collect raw material to create:
 - **Personas** (fictional user types)
 - **Profiles**
 - **Scenarios**
 - **Task descriptions**

Example:

You're designing a **hospital ward system**. You go and observe nurses during shifts:

- When do they check charts?
- How often are they interrupted?
- What shortcuts do they use?
- What frustrates them?

From this, you write scenarios like "Night nurse Ali checks patient vital signs while handling interruptions."

👉 **MEMORIZE:**

Ethnographic research = **observe users in their real context** to gather deep qualitative data for personas/scenarios.

3. Participatory design

Core idea:

Instead of designers working alone and then showing the product to users, in participatory design **users become part of the design team**.

- You **tap their knowledge, skills, and emotional reactions**.
- You run **workshops** where:
 - Users
 - Designers
 - Developersall work **together** on design ideas (e.g., sketching interfaces, rearranging screens).

Example:

Designing an educational app:

- Invite teachers and students.
- In a workshop, everyone draws what the “ideal lesson screen” should look like.
- They discuss why some layouts feel confusing or comforting.

👉 **MEMORIZE:**

Participatory design = **users on board with the design team**, via workshops, to co-create the design.

4. Focus group research

When used:

Early stages of a project, when concepts are still fuzzy.

What it is:

- A **group discussion** with representative users
- To evaluate **preliminary concepts** (not full products yet)
- Often like a structured brainstorming / “proof of concept preview”

Tools used:

- Paper & pencil drawings
- Storyboards
- Plastic models or mock-ups

Example:

You’re thinking of redesigning a ride-hailing app. In the focus group you:

- Show rough sketches of 3 booking flows
- Ask people what they like/dislike
- Listen to their concerns and preferences

👉 MEMORIZE:

Focus group = early-stage **group discussion** with representative users using rough artifacts (storyboards, sketches) for concept feedback.

5. Surveys

When used:

Initial stages of product implementation or exploration, when you want feedback from a **broad base of users**.

What it does:

- Checks **preferences** about existing or potential products
- Lets you reach **many people** online

Question types:

- **Open-ended** questions
 - User writes their own answer (e.g., “What do you like most about this app?”)
- **Closed-ended** questions
 - Multiple choice / Likert scale (e.g., “Rate satisfaction from 1 to 5”)

Surveys give mostly **quantitative data** (numbers, percentages), although open-ended can give text insights.

Example:

You email a questionnaire to 200 users asking:

- How often they use your mobile banking app
- What features they use
- How satisfied they are (1–7 scale)

6. Walkthroughs

Used at **prototype stage**.

Idea:

- You “walk through” a sequence of actions **as if a user is performing them**.
- Participants (or evaluators) imagine the user’s **route** through the product.

Uses:

- Explore how a user might **fare with an early concept** or prototype.
- Construct **work guides**: “If user wants X, they should do $Y \rightarrow Z \rightarrow \dots$ ”

Roles:

- Participants assume **specific roles**:
 - E.g., one person is “user”
 - Another is “system”
 - Another is “observer”

Example:

You have a prototype for an online exam system. You do a walkthrough:

1. User logs in
2. Selects exam
3. Starts test
4. Submits

At each step, you ask:

“Would a novice user know what to do next?”

“Are instructions clear?”

7. Open and closed card sorting

Goal: design “**findability**” – how easy it is to find information or functionality.

You give participants **cards**, each with a piece of content or function written on it (e.g., “Profile”, “Change password”, “Order history”).

7.1 Open card sorting

- You **don't** give category names.
- Participants group cards into piles and **invent their own category names**.

Example categories they might create:
“Account settings”, “Orders”, “Help”.

7.2 Closed card sorting

- You **do** give pre-defined categories.
- Participants sort cards into those categories.

Example:

Categories: “Account”, “Shop”, “Help”

Users assign each card to an existing category.

👉 MEMORIZE:

Card sorting = designing navigation/findability.

Open → users create category names.

Closed → users sort into **your** predefined categories.

8. Paper prototyping

What happens:

- Show users **paper versions** of screens (hand-drawn or printed).
- Ask them questions and see how they think they would use it.

Sometimes:

- “Expert mockup responses” are sketched on graph paper (e.g., “if user clicks here, show this screen”).

Why it's useful:

- Very **cheap and fast** to change.

- Great for **early feedback** before coding.

Example:

You draw a phone app on A4 sheets:

- Home screen
- Profile screen
- Cart screen

You simulate transitions by flipping papers when the user “clicks” a button.

9. Expert / Heuristic evaluations

What it is:

- **Experts** review a system or product (not just normal users).
- Often they are “**double specialists**”:
 - Know HCI/usability principles
 - Know the domain (e.g., medical, finance)

They use **heuristics** (rules-of-thumb) like:

- Visibility of system status
- Match with real-world
- Consistency
- Error prevention, etc.

They point out issues like:

- Confusing labels
- Inconsistent buttons
- Missing feedback

Example:

You hire a UX expert to review a hospital dashboard. She systematically points out:

- “This alarm icon is not visible enough”
 - “Error messages are vague”
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10. Usability testing – main idea

Usability testing = techniques to collect **empirical data** while observing **real end users** using the product.

Types:

- Can be **formal tests** (true experiments) → confirm/refute a **specific hypothesis**
- Or **iterative cycles of tests**:
 - Expose usability problems
 - Fix them
 - Test again
 - Slowly shape the product

👉 **MEMORIZE:**

Usability testing = observing real users doing real tasks, collecting data (time, errors, satisfaction) to evaluate and improve usability.

11. Why usability testing? (Goals)

Three main reasons (goals):

1. Informing design

- Understand how people actually use it
- See where they get stuck, what confuses them

2. Eliminating design problems & frustration

- Find painful spots before launch
- Reduce rage-clicks, confusion, errors

3. Improving profitability

- Better usability → more users complete tasks (buy products, subscribe)
- Less call-center support and fewer complaints

👉 **MEMORIZE** these 3 goals.

12. “Classical approach” methodology vs modern usability testing

Your slide gives the **classical experimental methodology**:

- **Hypothesis must be formulated**
- Participants randomly assigned to experimental conditions
- Tight controls
- Control groups
- Sample size big enough for **statistically significant differences**

This is like what you saw in Week 8 (true experiments).

But for **usability testing** in practice, the **basic elements** differ a bit.

13. Basic elements of usability testing (as in your slide)

Instead of strict hypotheses, we usually have **research questions or test objectives**, e.g.:

- “Can users complete a money transfer in less than 3 minutes?”
- “Where do users get stuck while booking a ride?”

Other basic elements:

1. **Research questions/test objectives**, not necessarily full statistical hypotheses.
2. **Representative sample of end users**
 - May or may not be randomly chosen (often we recruit target users, not just random people).
3. **Representation of the actual work environment**
 - Try to simulate realistic context (devices, tasks, noise).
4. **Observation of users**
 - They use or review the product (or prototype) while you watch.
5. **Controlled interviewing and probing by the moderator**
 - Ask “Why did you click that?”, “What were you expecting?”
6. **Collect both quantitative and qualitative data**
 - Quantitative: time, error rate, success rate
 - Qualitative: comments, think-aloud verbalizations
7. **Recommendations for design improvements**
 - At the end, you suggest what to fix and how.

👉 **MEMORIZE** this list (or at least 4–5 key items).

14. Limitations of usability testing

Your slide clearly lists:

1. **Testing is always an artificial situation**
 - Lab ≠ real life (people may behave differently under observation).
2. **Results do not prove a product works**
 - A few tests that go well don't mean it will be perfect for all users.
3. **Participants are rarely fully representative**
 - Hard to get a perfectly representative group of all target users.
4. **Testing is not always the best technique**
 - Sometimes ethnography, surveys, or expert reviews are better at a certain stage.

👉 **MEMORIZE** these 4 points.

15. Types of usability tests

Your slides list 4 types (plus “comparison”):

- **Exploratory (formative)**
- **Assessment (summative)**
- **Validation (verification)**
- **Comparison** (linked with the above three)

We'll go one by one.

16. Exploratory or Formative study

When:

Quite **early in the development cycle**.

Objective:

Examine the **effectiveness of the preliminary design** – at a high level.

Example from slide:

- App designer wants to see how well the interface:
 - Supports users' tasks within a goal
 - Communicates intended workflow
 - Allows navigation between screens and within a screen

16.1 Perspective: task-oriented user view (II)

They look at:

- Overall organization of subject matter (is content grouped logically?)
- Use of graphics vs verbal text
- How well the format supports **findability**
- Where users might need **assistance and messages**
- How to present **reference information**

16.2 Typical user-oriented questions (III)

Examples from slides:

- What do users **think** about using the product?
- Does the product's **basic functionality have value**?

- How easily do users make **inferences about how to use** it, using their previous experience?
- What **prerequisite information** is required to use it?
- Which functions are “**walk up and use**” (immediate) and which need documentation/help?
- How should the **table of contents** be organized for both novices and experts?

16.3 Methodology (IV)

- Use **prototypes**:
 - **Horizontal representation**: many features, shallow depth (all menus visible, but not fully functional)
 - **Vertical representation**: deep flow for a few features (full detail of one workflow)

👉 MEMORIZE:

Exploratory/formative = early, high-level design, questions about basic value, workflow, navigation, and “walk up and use” functions.

17. Assessment or Summative test

When:

Either **early or midway** in the product development cycle, usually **after fundamental/high-level design** is done.

Objective:

To **expand findings** of exploratory tests by evaluating usability of **lower-level operations and aspects**.

Example from slide:

- How well can a user **actually perform a full realistic task** (end-to-end)?

17.1 Methodology (II)

- User **performs tasks** (not just walkthrough/commenting).
- Test moderator **reduces interaction** with participants (less thinking-aloud, more actual behavior).
- **Quantitative measures** collected:
 - Task time
 - Number of errors
 - Success rates

👉 **MEMORIZE:**

Assessment/summative = after basic design, focus on actual task performance with more **quantitative data** and less moderator interaction.

18. Validation or Verification test

When:

Late in the **deployment cycle**.

Objective:

- Measure usability of product against **established benchmarks** (standards).
- For **verification** tests:
 - Confirm that problems found earlier have been fixed
 - Ensure **no new issues** were introduced

Example:

Compare product's performance against **previous release**, or against **industry benchmarks**.

18.1 Methodology (II)

- Before test, **standards/benchmarks** for tasks are defined.

- Participants perform tasks with **minimal or no moderator interaction**.
- Central focus: collection of **quantitative data** (time, success, error).
- But you also identify reasons for **poor performance** if metrics are below benchmarks.

👉 **MEMORIZE:**

Validation/verification = late stage, measure against **benchmarks**, low moderator interaction, strongly quantitative.

19. Comparison test

When:

No fixed point in lifecycle – can be used **with any of the above three**.

Objective:

- Compare **two or more designs**
 - Two interface styles
 - Old manual vs new manual
 - Two navigation structures, etc.

19.1 Methodology (II)

- Design team is **forced to stretch their ideas** – they don't just keep doing the same pattern.
- Participants are forced to think **why one design is better** and which aspects make it better.

Example:

You show Design A and Design B of an onboarding flow:

- Ask users to use both

- Compare metrics (time, satisfaction)
 - Ask them to explain their preferences
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3. Practice Questions

Now MCQs, short answer, and one case study based **only on Week 9**.

A. MCQs (with answers + explanations)

MCQ 1

Which of the following best describes **ethnographic research** in HCI?

- A. Inviting users to the lab and asking them to fill a questionnaire
- B. Observing users in their **natural environment** while they perform real tasks
- C. Experts reviewing a design using usability heuristics
- D. Conducting a controlled experiment with random assignment

Correct answer: B

Explanation:

Ethnographic research is a **qualitative** method where you observe users **where they would use the product**, to gather data about users, their tasks, and goals.

MCQ 2

In **open card sorting**, participants:

- A. Sort cards into categories that the researcher provides
- B. Sort cards randomly to see which ones they like
- C. Create their **own category names** and group cards into them
- D. Only rank cards in order of importance

Correct answer: C

Explanation:

Open card sorting = participants are given content cards without existing category names and are asked to **name and group** them into categories themselves. Closed card sorting uses pre-existing categories.

MCQ 3

Which usability test type is **quite early** in the development cycle and focuses on examining the **effectiveness of preliminary design** and workflow?

- A. Assessment (summative) test
- B. Exploratory (formative) study
- C. Validation (verification) test
- D. Comparison test

Correct answer: B

Explanation:

Exploratory/formative studies are conducted early to evaluate **preliminary designs**, understand workflows, navigation, and basic value of the product.

MCQ 4

Which of the following is **NOT** listed as a limitation of usability testing?

- A. Testing is always an artificial situation
- B. Test results do not prove that a product works
- C. Participants are rarely fully representative of the target population
- D. Testing replaces the need for all other research methods

Correct answer: D

Explanation:

Your slide explicitly says that **testing is not always the best technique** – meaning it does **not** replace other methods. The first three options are directly listed as limitations.

MCQ 5

In a **validation/verification test**, which of the following is most central?

- A. Collecting open-ended comments only
- B. Measuring usability against **predefined benchmarks**
- C. Brainstorming new design ideas with users
- D. Creating personas and scenarios

Correct answer: B

Explanation:

Validation/verification tests are carried out **late** in the development cycle to measure usability against **established benchmarks** and confirm that earlier problems are fixed.

B. Short Answer Questions (with model answers)

SAQ 1

What is participatory design, and how is it different from just “asking for feedback” at the end?

Model answer:

Participatory design is when **users are on board with the design team**, and they actively **co-create** the design in workshops, contributing their knowledge, skills, and emotional reactions. This is different from simply asking for feedback at the end because users are involved **throughout the design process**, shaping early ideas rather than just reacting to a finished product.

SAQ 2

Define “heuristic evaluation” and explain who typically conducts it.

Model answer:

Heuristic evaluation is a review of a product or system by **experts**, who check it against known usability principles (heuristics). The evaluators are often “**double specialists**” who understand both usability/interaction design and the specific application domain. They identify usability problems without involving end users directly.

SAQ 3

List three basic elements of usability testing according to your slides.

Model answer (pick any three):

1. Development of **research questions/test objectives**, rather than a strict hypothesis.
 2. Use of a **representative sample** of end users.
 3. Representation of the **actual work environment**.
 4. **Observation** of end users using or reviewing the product.
 5. **Controlled interviewing and probing** by the test moderator.
 6. Collection of **quantitative and qualitative measures**.
 7. Providing **recommendations for design improvements**.
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SAQ 4

Differentiate between an exploratory (formative) study and an assessment (summative) test.

Model answer:

An exploratory/formative study happens **early** in the development cycle and focuses on evaluating **preliminary design**, overall organization, workflow, and whether the basic functionality has value. An assessment/summative test is conducted **after** the high-level design is established and focuses on how well users can **actually perform realistic tasks**, with more emphasis on **quantitative measures** like time and error rates.

SAQ 5

What are two limitations of usability testing mentioned in your slides?

Model answer (any two):

1. Testing is always an **artificial situation**, not exactly like real life.
2. Test results do **not prove** that a product works for all users or all scenarios.
3. Participants are rarely **fully representative** of the target population.

4. Testing is **not always the best technique** to use in every situation.
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C. Case Study (with full answer)

Case Study

You are designing a **new mobile banking app**. The bank wants to:

- Understand if users **like** the overall concept and structure.
- Evaluate how well users can **perform complete tasks** (like sending money).
- Check, near launch, whether the app **meets certain usability benchmarks** (e.g., transfer must be completed within 2 minutes by 90% of users).

Question:

1. Which **types of tests** (from Week 9) would you use at each of these three stages:
 - Early concept feedback
 - Mid-development usability evaluation
 - Near launch benchmark checking
 2. For each test type you choose, briefly state:
 - **When** it is used in the lifecycle
 - **Objective**
 - One key aspect of its **methodology** based on your slides
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Model Answer

1. **Early concept feedback**

- **Test type: Exploratory (formative) study**
- **When:** Quite **early** in the development cycle, when only preliminary designs exist.
- **Objective:**
 - Examine how well early designs support users' **tasks and goals**, whether the **workflow is clear**, and whether users see **value** in the basic functionality.
- **Methodology point:**
 - Use **prototypes** (horizontal or vertical) and investigate issues like overall organization, navigation, and what is “walk up and use”. Users are asked questions such as “How do you think you would transfer money from here?”

2. Mid-development usability evaluation

- **Test type: Assessment (summative) test**
- **When:** Early or **midway** into development, after high-level design is done.
- **Objective:**
 - Expand early findings by evaluating usability of **lower-level operations** and **full realistic tasks**, such as “log in → check balance → send money → confirm transaction”.
- **Methodology point:**
 - Users **perform tasks** (not just walkthroughs), the moderator **reduces interaction**, and you collect **quantitative measures** such as task completion time, errors, and success rates.

3. Near launch benchmark checking

- **Test type: Validation/Verification test** (possibly with a **Comparison test** if comparing against old app).
- **When:** **Late** in the deployment cycle, close to launch.

- **Objective:**
 - Measure usability against predefined **benchmarks** (e.g., “90% of users complete a transfer within 2 minutes”) and verify that previously discovered problems have been fixed and no new ones introduced.
- **Methodology point:**
 - Before testing, set **standards** (time targets, success rates); let users perform tasks with **minimal moderator interaction**; focus on **quantitative data** (times, success rates, error counts) and then investigate reasons for any substandard performance.

If the bank also wants to compare **two design versions** of the app (e.g., old vs new), you would add a **Comparison test**, where users try both versions and you collect metrics and ask them which one is better and why.